# Computer Programming Languages

Lecture – 1

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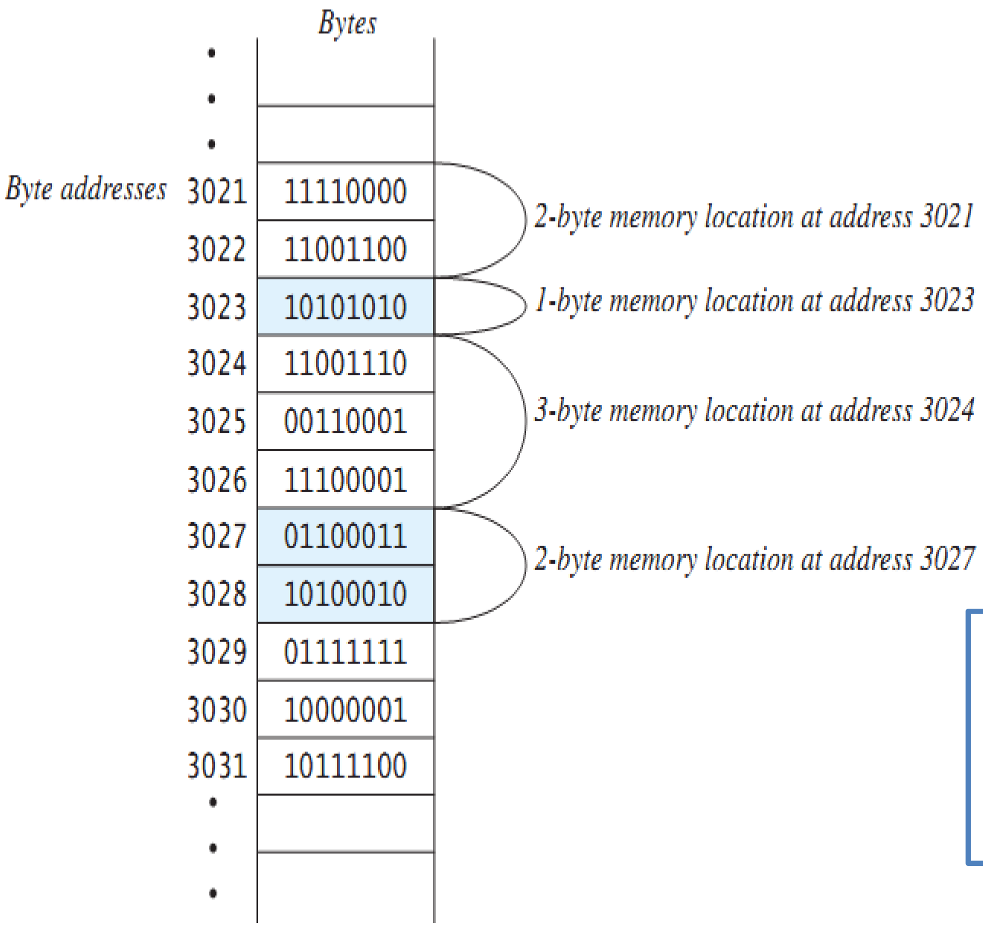
# Computer Basics

- Computer systems consist of <u>hardware</u> and <u>software</u>
- A set of instructions for the computer to carry out is called a <u>program</u>
- All the different kinds of programs used to give instructions to the computer are <u>collectively</u> referred to as <u>software</u>

# Hardware and Memory

- The CPU, or central processing unit, or processor, performs the instructions in a program
- Computer's memory holds data for the computer to process
  - It holds the result of the computer's intermediate calculations
  - A byte is the quantity of memory
    - 1 gigabyte of RAM is approximately 1 billion bytes of memory

# Hardware and Memory




Main memory consists of a long list of numbered bytes

The number of a byte is called its address

A byte is the smallest addressable unit of memory

Data (numbers, letters and string of characters) is encoded as a series of 0s and 1s and placed in the computer's memory

# Programming Languages

- Computer programs(software) are instructions that tell a computer what to do
- Many programming languages
  - All programs are converted into instructions that computer can execute
- Types of Languages
  - Machine Language
  - Assembly Language
  - High-Level Language

**Example**

**To add Two Numbers:**

1101101010011010

add 2, 3, result

result = 2 + 3

| *Language* | *Description* |
|---|---|
| Ada | Named for Ada Lovelace, who worked on mechanical general-purpose computers. The Ada language was developed for the Department of Defense and is used mainly in defense projects. |
| BASIC | Beginner's All-purpose Symbolic Instruction Code. It was designed to be learned and used easily by beginners. |
| C | Developed at Bell Laboratories. C combines the power of an assembly language with the ease of use and portability of a high-level language. |
| C++ | C++ is an object-oriented language, based on C. |
| C# | Pronounced "C Sharp." It is a hybrid of Java and C++ and was developed by Microsoft. |
| COBOL | COmmon Business Oriented Language. Used for business applications. |
| FORTRAN | FORmula TRANslation. Popular for scientific and mathematical applications. |
| Java | Developed by Sun Microsystems, now part of Oracle. It is widely used for developing platform-independent Internet applications. |
| Pascal | Named for Blaise Pascal, who pioneered calculating machines in the seventeenth century. It is a simple, structured, general-purpose language primarily for teaching programming. |
| Python | A simple general-purpose scripting language good for writing short programs. |
| Visual Basic | Visual Basic was developed by Microsoft and it enables the programmers to rapidly develop graphical user interfaces. |

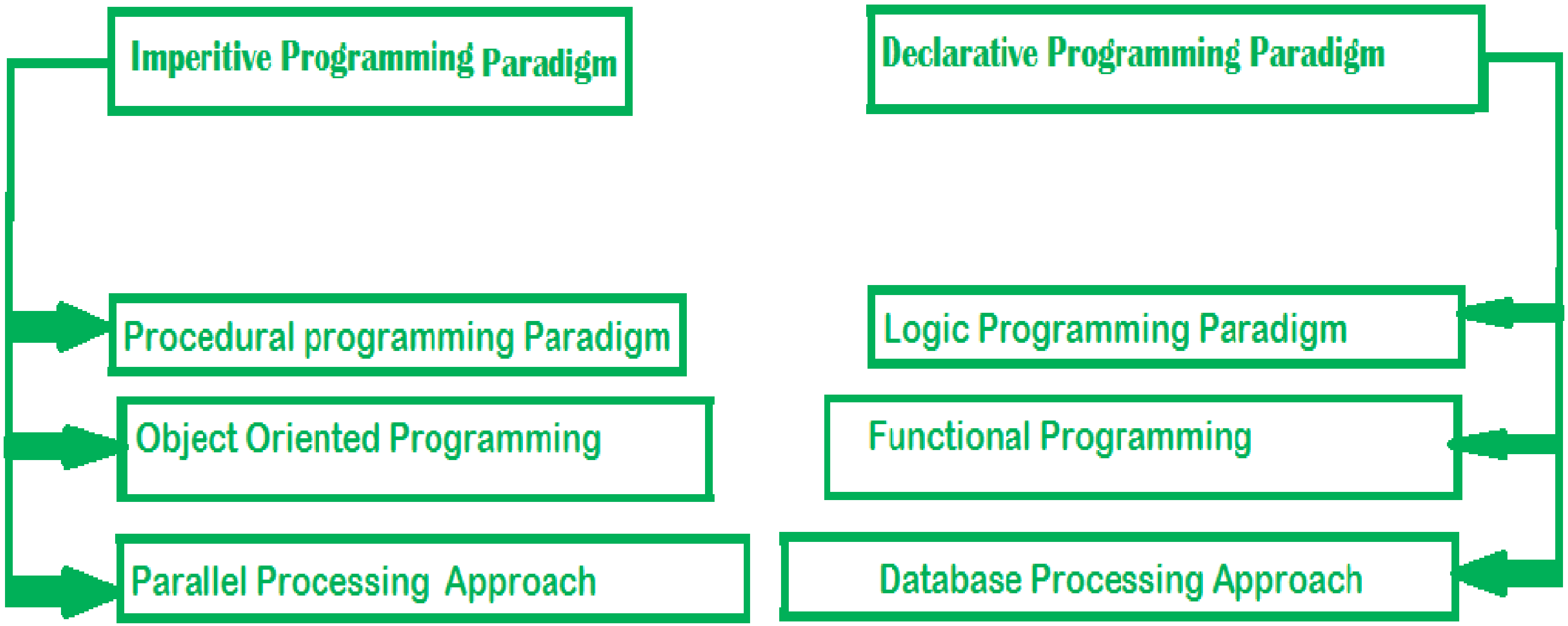
# Programming Paradigm

- A **paradigm** is a system of concepts and practices that reflect the current state of our understanding of the field
- A **programming paradigm** refers to a style, way, or classification of programming
- **Imperative** in which the programmer instructs the machine *how to change its state.*

  **Examples:** C, C++, Java, PHP, Python, Ruby etc.
- **Declarative** the programmer states only *what the result should look like*, **not** how to obtain it

```
select upper(name)
from people
where length(name) > 5
order by name
```

# Programming Paradigms



# Programming Paradigm

- **Programming languages** are used in order to solve problems
  - Structured/Procedural Programming
  - Functional Programming
  - Object-oriented Programming

# Structured Programming

- Kind of imperative programming
- Control flow is defined by nested loops, conditionals, and subroutines
- *Sequence*, *Selection* and *Repetition*

```
x= 5;
y = 11;
z = x + y;
System.out.println(z);
```

```
if (x % 2 == 0)
{
   System.out.println("The
number is even.");
}
```

```
while(x < 100)
{
   System.out.println(x);
   x = x * x
}
```

# Procedural Programming

- A language derived from structure programming and based on the use of call procedures
- **Procedures** are the functions, routines, or subroutines that specify the computational steps that need to be performed
- A program is designed using a top-down approach in procedural programming
- **Examples**: COBOL, BASIC, PASCAL, FORTRAN, and C

# Example (C Language)

```c
#include <stdio.h>

// Function to add two numbers
int add(int a, int b) {
    return a + b;
}

int main() {
    int num1, num2;
    printf("Enter two numbers: ");
    scanf("%d %d", &num1, &num2);
    int sum = add(num1, num2);
    printf("Sum: %d\n", sum);

    return 0;
}
```

# Object Oriented Programming (OOP)

- Object-oriented programming is about creating objects that contain both **data** and **methods**
- **Classes** and **objects** are the two main aspects of object-oriented programming

| Class | Object |
|---|---|
| Fruit | Apple, Banana, Mango |
| Car | Volvo, Audi, Toyota |
| Student | Ali, Hamza, Ahmad |

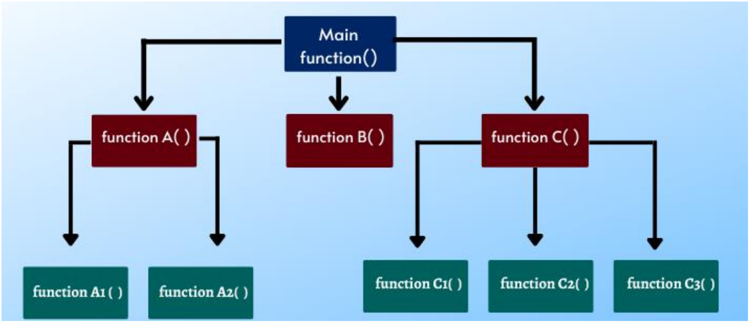
- **Examples:** C++, C#, Java, Python.

# Programming Paradigm

## Example

```
public class Car {
  String carReg;
  int carModel;
  String carClass;
  public Car(String num) {
    carReg = num;
  }
  public String getReg(){
       return carReg;
  }
  public static void main(String[] args) {
    Car Audi = new Car("AX-7865");
    System.out.println(Audi.getReg());
  }
}
```

# Functional Programming (FP)




- Executing a series of mathematical functions is the key principle of this FP Paradigm
- In contrast to an imperative style which emphasizes "**how to solve**", this style focuses on "**what to solve**"
- It uses **expressions** rather than a **statement**
  - Expressions are evaluated to produce values, while statements are executed to assign variables.
- **Examples:** C++, Python, Java Script, Lisp, Haskell

# FP-Example

- **Example**
  - Summing the integers 1 to 10 in Java:
  - The computation method is <u>variable assignment</u>

```
int total = 0;
for (int i = 1; i ≤ 10; i++)
   total = total + i;
```

- Example
  - Summing the integers 1 to 10 in Haskell (Functional Language) :

```
sum [1..10]
```

  - The computation method is function application

# FP-Example

```
import java.util.Arrays;
import java.util.List;
public class Functional{
        public static void main(String[] args){
            List<Integer> numbers = Arrays.asList(11, 22, 33, 44, 55, 66, 77, 88, 99, 100);
                final int factor = 2;
//Declarative
                numbers.stream()
                .filter(number->number%2==0)
                .forEach(System.out::println);

//imperative
                for(int number: numbers){
                        if(number%2 ==0)
                                System.out.println(number);
                }
        }
}
```

# Program Development

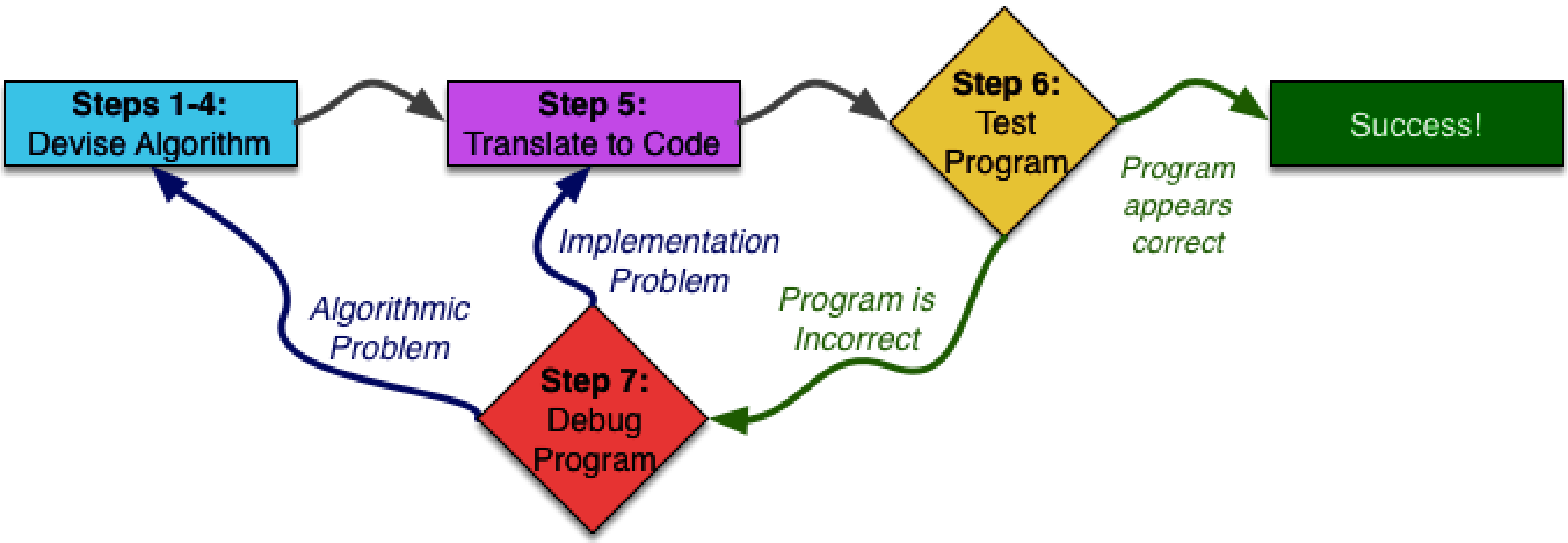
# Outline

- Program Development Cycle
- Algorithm
- The First Four Steps

# Programming: Plan First, Then Code

- A good programmer will plan first and write second
  - Breaking down a large programming task into several smaller tasks

# An Overview of Seven Steps



# Algorithm

- Set of well-defined logical steps that must be taken to perform a task
- ***For example****, suppose you have been asked to write a program to calculate and display the gross pay for an hourly paid employee.*

**Here are the steps that you would take:**

1. Get the number of hours worked.
2. Get the hourly pay rate.
3. Multiply the number of hours worked by the hourly pay rate.
4. Display the result of the calculation that was performed in Step 3.

# Pseudocode

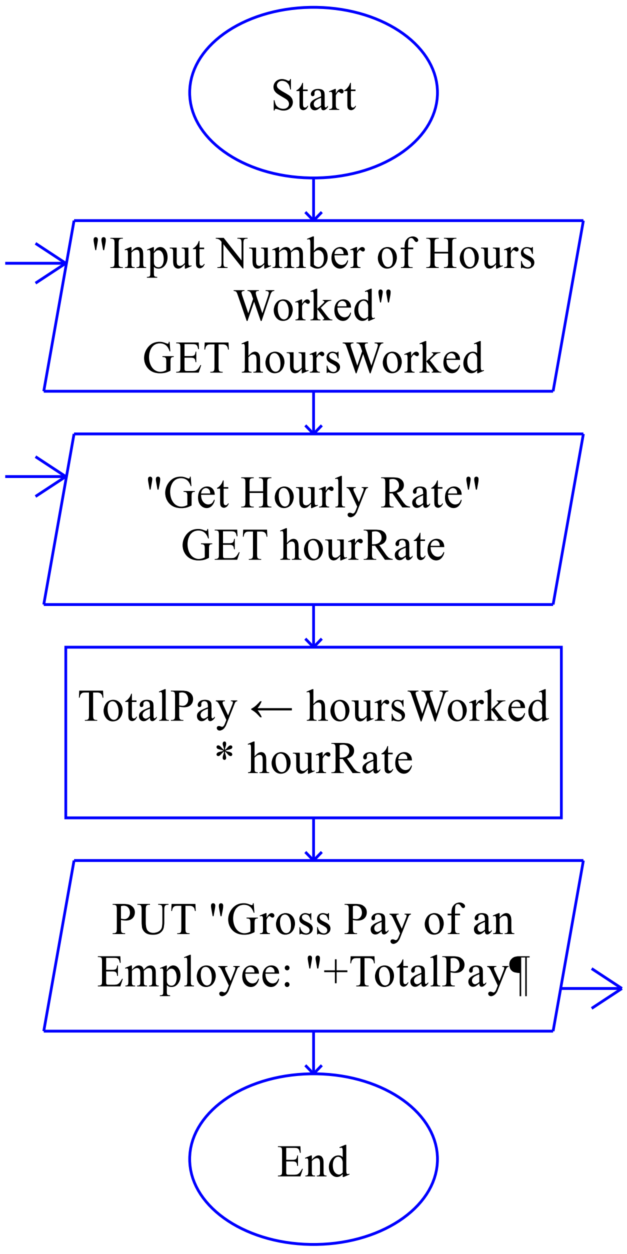
- Pseudocode: fake code
  - Informal language that has no syntax rule
  - Not meant to be compiled or executed
  - Used to create model program
    - No need to worry about syntax errors, can focus on program's design
    - Can be translated directly into actual code in any programming language

```
Display "Enter the number of hours the employee worked."
Input hours
Display "Enter the employee's hourly pay rate."
Input payRate
Set grossPay = hours * payRate
Display "The employee's gross pay is $", grossPay
```

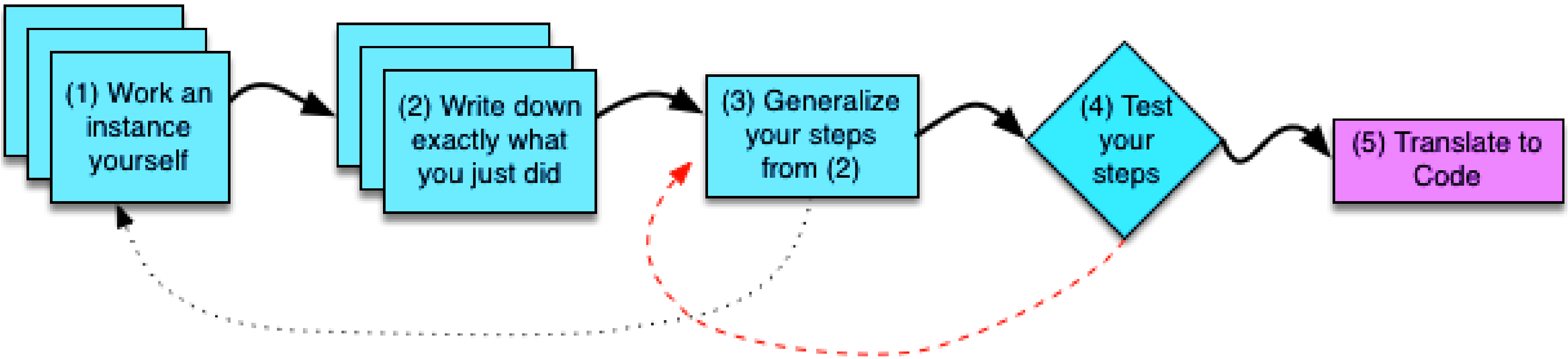
# Flowcharts

- <u>Flowchart</u>: diagram that graphically depicts the steps in a program
  - Ovals are terminal symbols
  - Parallelograms are input and output symbols
  - Rectangles are processing symbols
  - Symbols are connected by arrows that represent the flow of the program

# Flow Chart – Raptor Example



# First Four Steps




Step 1: Work an Example Yourself

Step 2: Write Down What You Just Did

Step 3: Generalize Your Steps

Step 4: Test Your Algorithm

# Example-1

Write a program to compute $x$ raised to the $y$ power ($x^y$)

**Step 1: Work an Example Yourself**

*x* = 3 and *y* = 4, getting an answer of 3^4 = 81.

# Example-1

**Step 2: Write Down What You Just Did**

```
Multiply 3 by 3
  You get 9
Multiply 3 by 9
  You get 27
Multiply 3 by 27
  You get 81
81 is your answer.
```

# Example-1

## Step 3: Generalize Your Steps

1-Replacing this occurrence of 3 with x

```
Multiply 3 by 3
   You get 9
Multiply 3 by 9
   You get 27
Multiply 3 by 27
   You get 81
81 is your answer.
```

```
Multiply x by 3
   You get 9
Multiply x by 9
   You get 27
Multiply x by 27
   You get 81
81 is your answer.
```

# Example-1

## Step 3: Generalize Your Steps

2-Find repetition

```
Start with n = 3
n = Multiply x by n
n = Multiply x by n
n = Multiply x by n
n is your answer.
```

```
Start with n = 3
Count up from 1 to y-1 (inclusive), for each number you count,
  n = Multiply x by n
n is your answer.
```

# Example-1

## Step 3: Generalize Your Steps

```
Start with n = x
Count up from 1 to y-1 (inclusive), for each number you count,
  n = Multiply x by n
n is your answer.
```

# Example-1

- Step 4: Test Your Algorithm

```
If y is 0 then
  1 is your answer
Otherwise:
  Start with n = x
  Count up from 1 to y-1 (inclusive), for each number you count,
    n = Multiply x by n
  n is your answer.
```

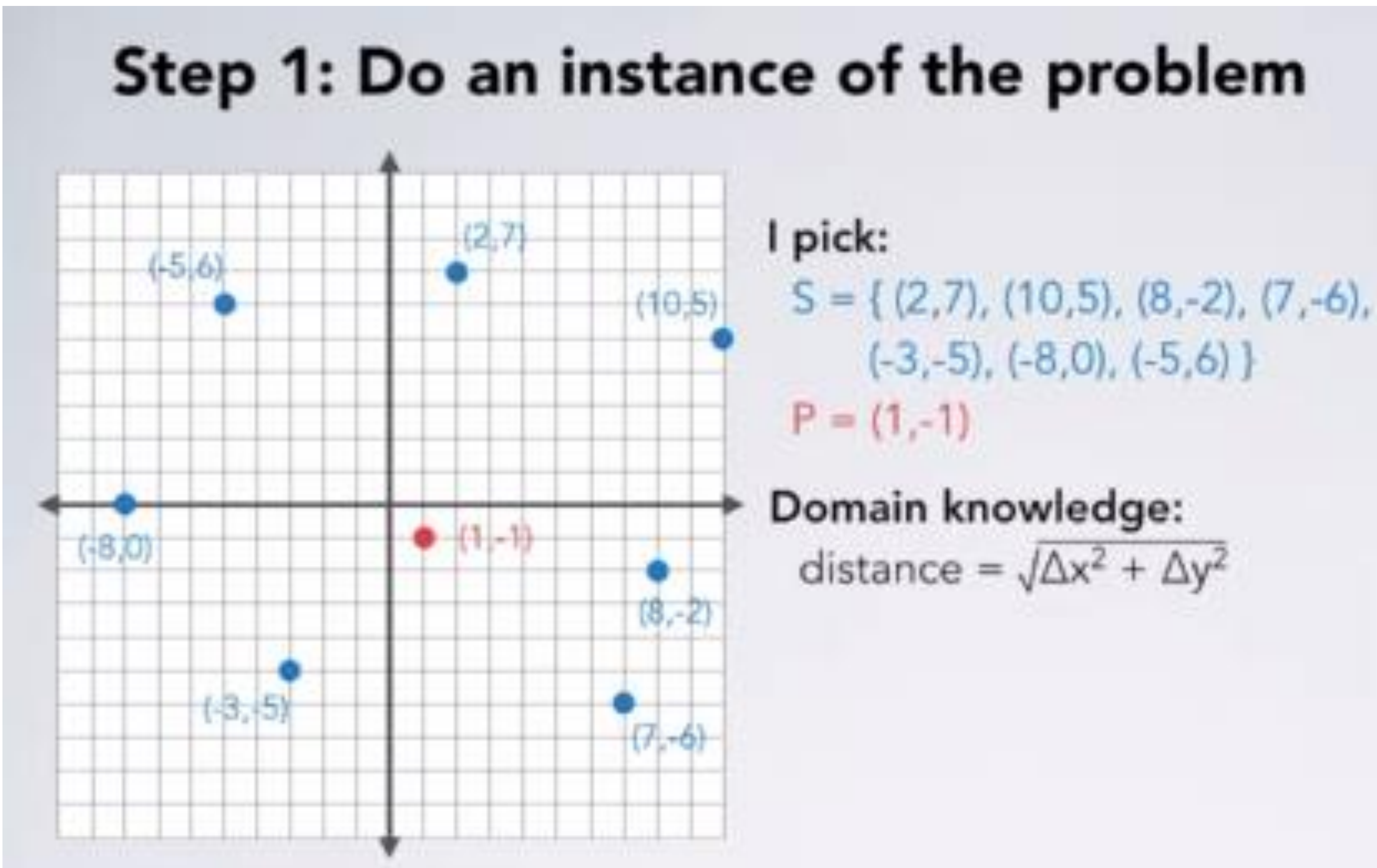
# Example-1

- Step 4: Test Your Algorithm

```
Start with n = 1
Count up from 1 to y (inclusive), for each number you count,
  n = Multiply x by n
n is your answer.
```

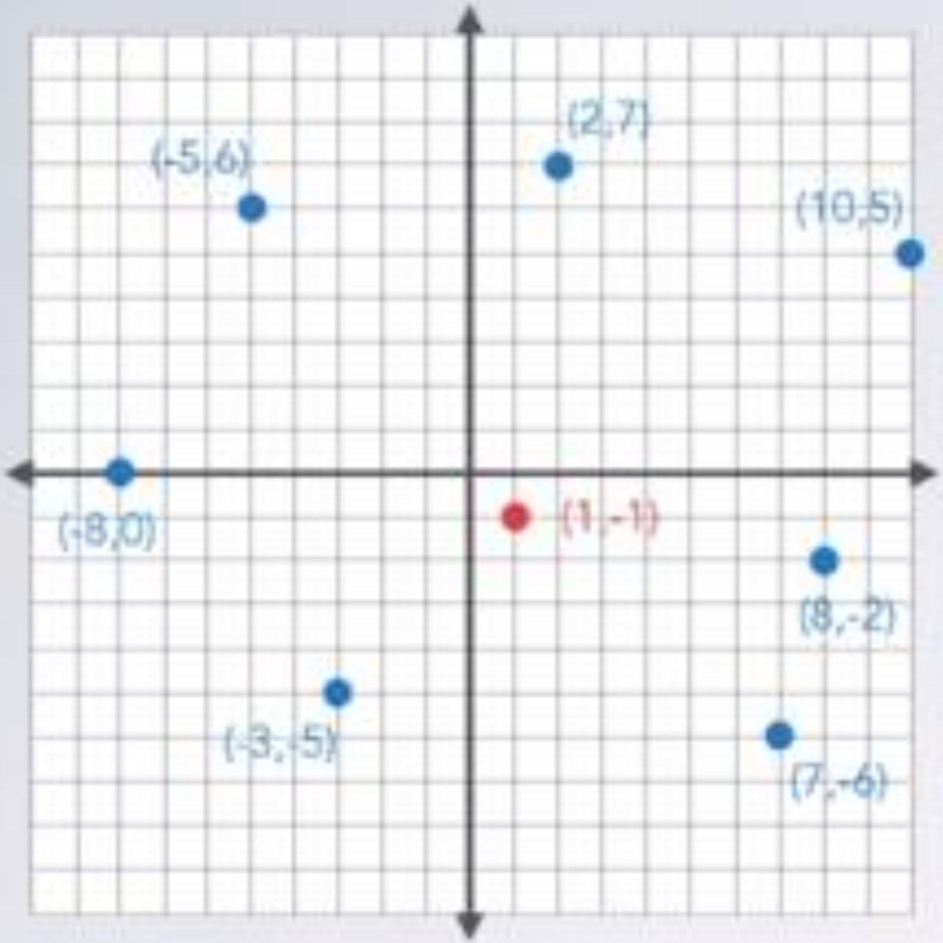
# Example-2

- **Problem:** Find the closest point out of a set of points to some other given point



# Example-2

## Step 1: Do an instance of the problem




**I pick:**

S = { (2,7), (10,5), (8,-2), (7,-6), (-3,-5), (-8,0), (-5,6) }

P = (1,-1)

**Domain knowledge:**

$$\text{distance} = \sqrt{\Delta x^2 + \Delta y^2}$$

# Example-2

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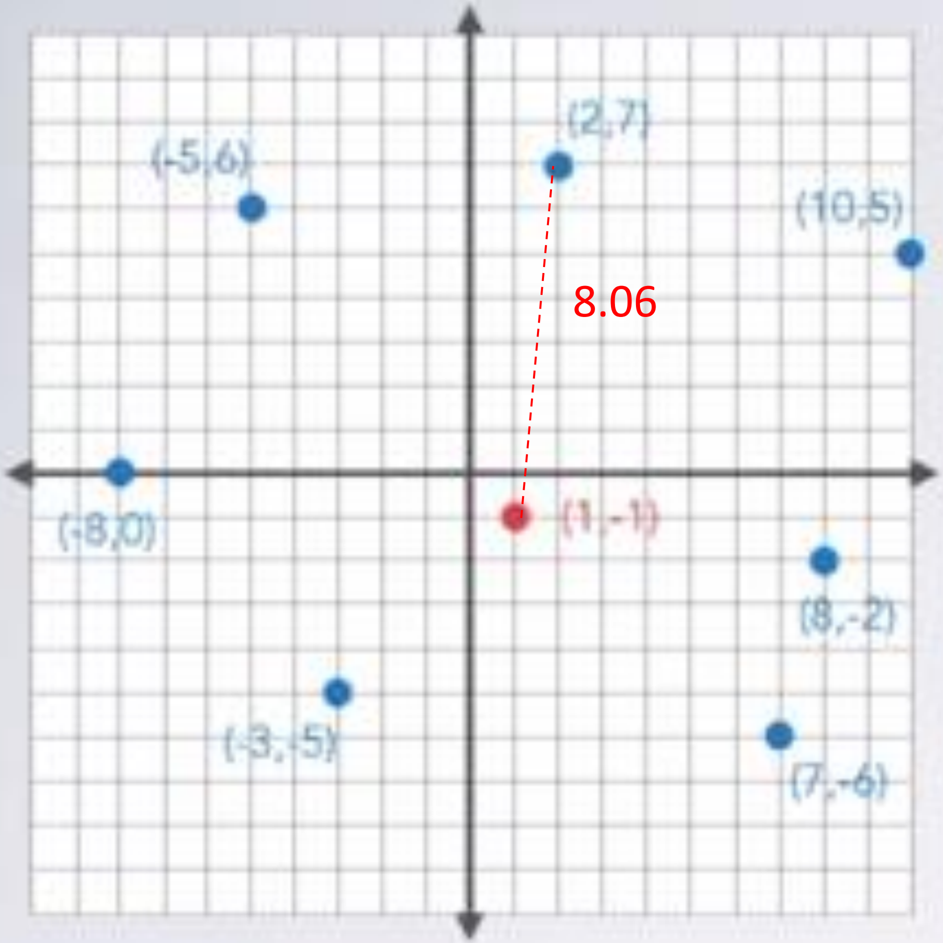


**I pick:**

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**Smallest Distance: 8.06**

# Example-2

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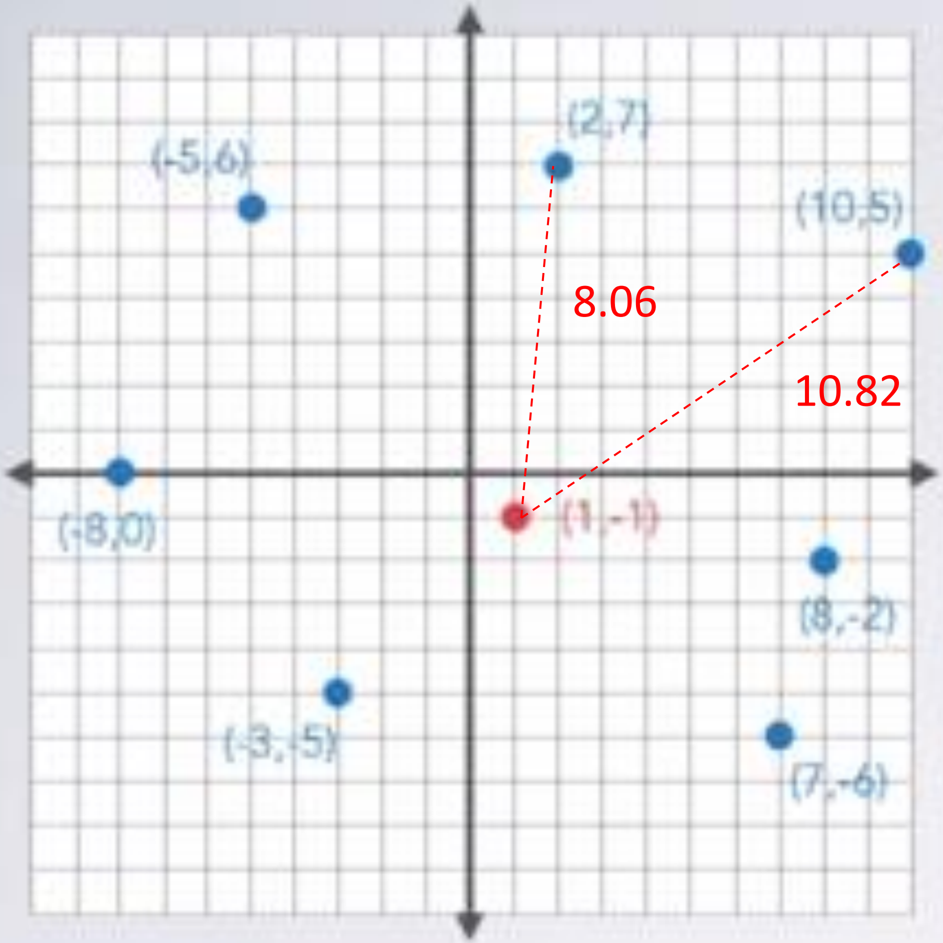


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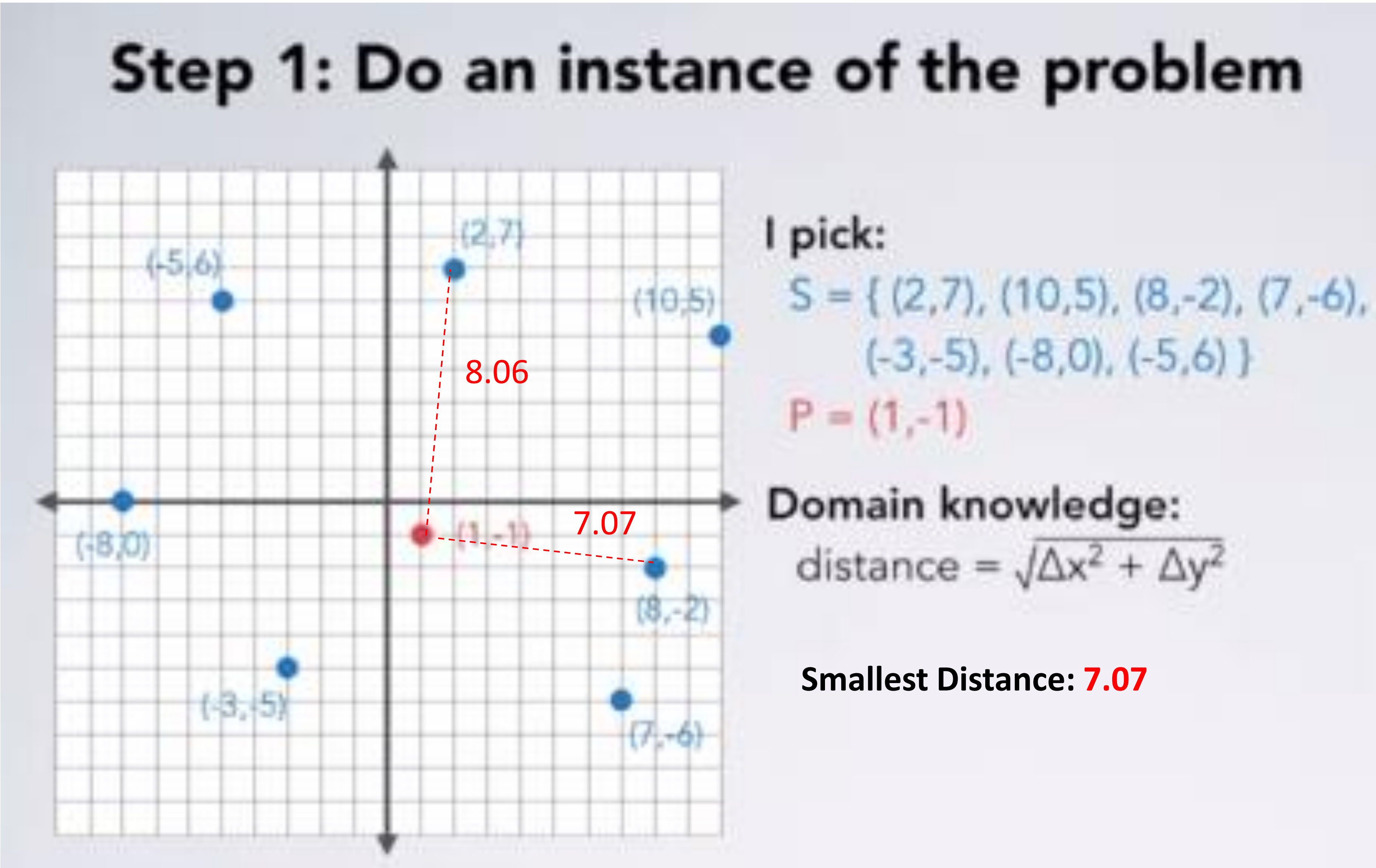
**Smallest Distance: 8.06**

# Example-2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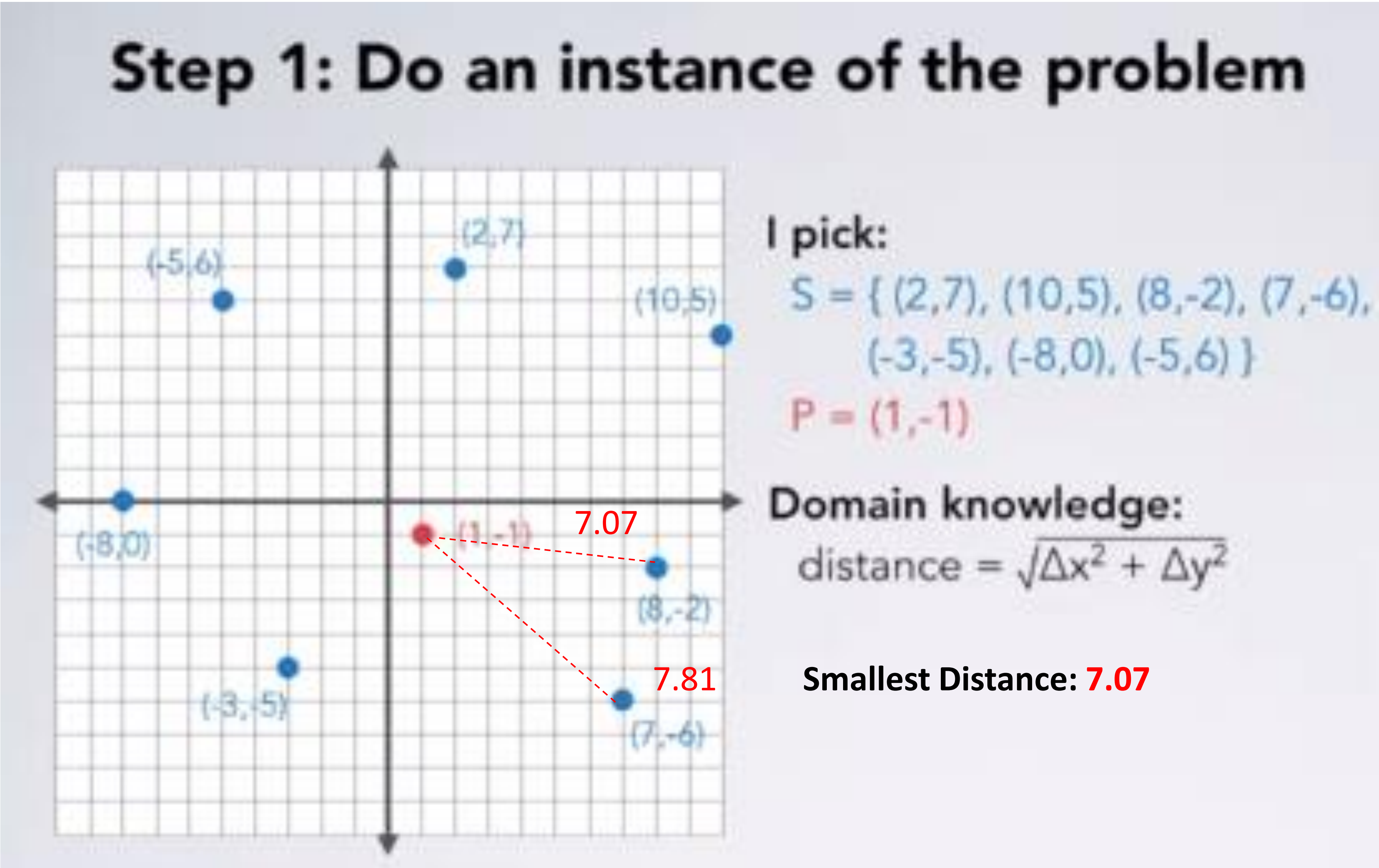
Smallest Distance: 7.07

# Example-2



Step 1: Do an instance of the problem
(2,7)
(-5,6)
(10,5)
(-8,0)
(1,-1)
7.07
(8,-2)
7.81
(-3,-5)
(7,-6)
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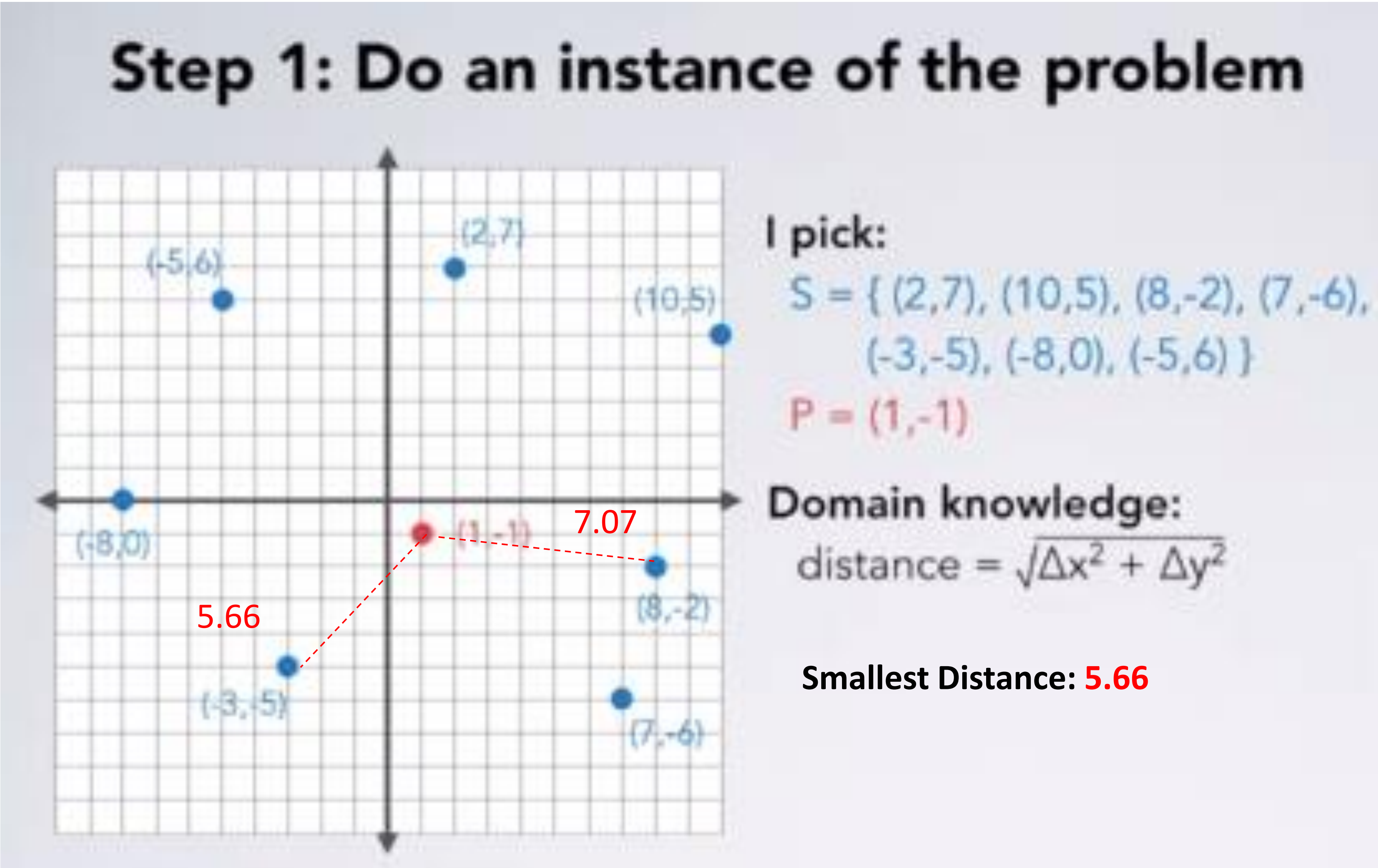
# Example-2


Step 1: Do an instance of the problem
(2,7)
(-5,6)
(10,5)
(-8,0)
(1,-1)
7.07
(8,-2)
5.66
(-3,-5)
(7,-6)
I pick:
S = { (2,7), (10,5), (8,-2), (7,-6), (-3,-5), (-8,0), (-5,6) }
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Domain knowledge:
distance = $\sqrt{\Delta x^2 + \Delta y^2}$
Smallest Distance: 5.66

# Example-2

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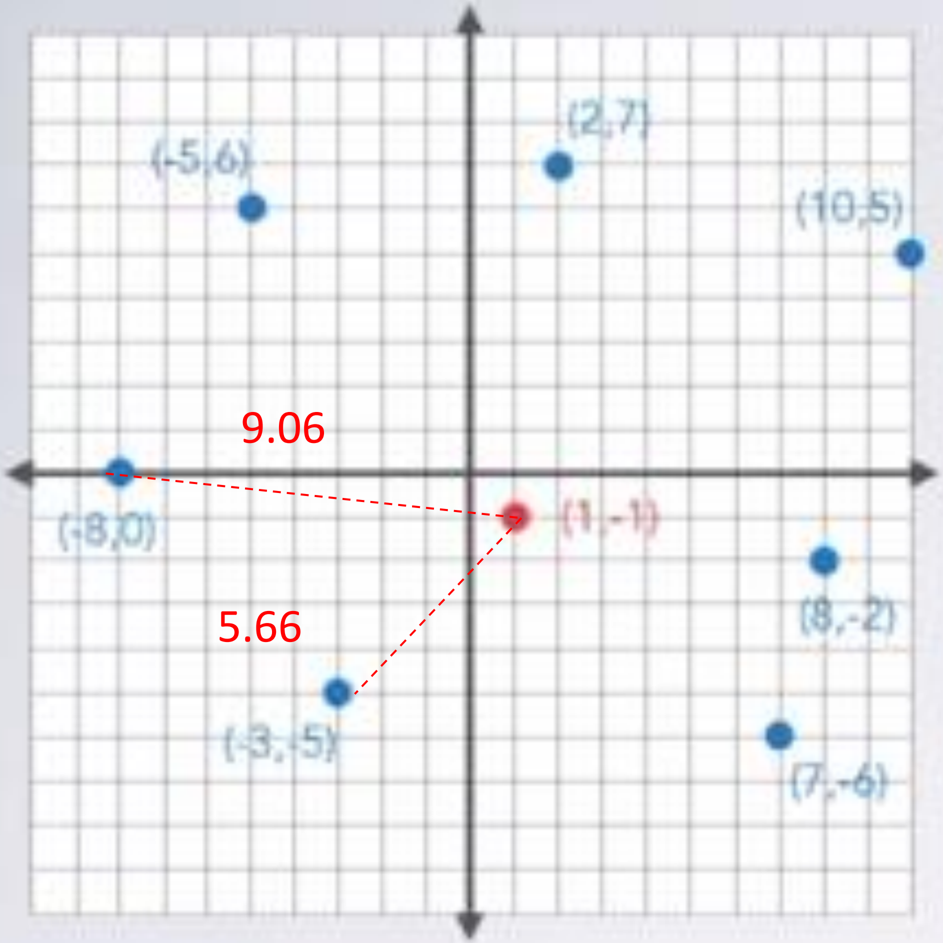


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$$\text{distance} = \sqrt{\Delta x^2 + \Delta y^2}$$

**Smallest Distance: 5.66**

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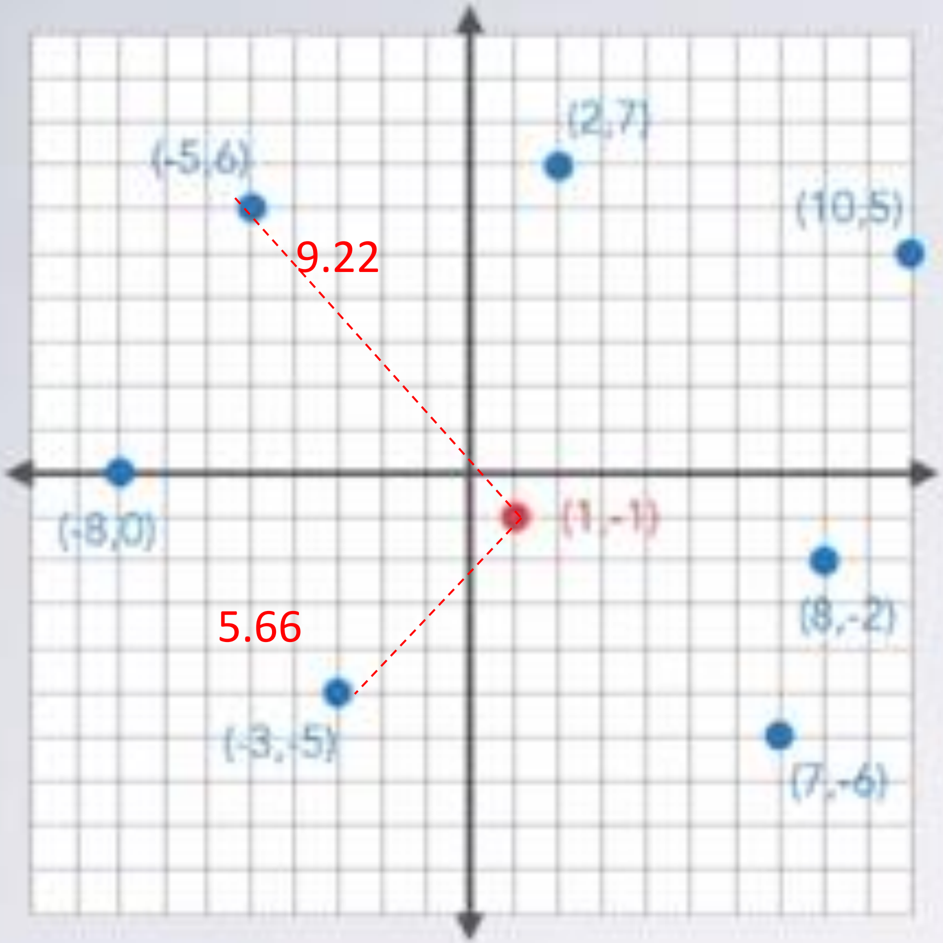


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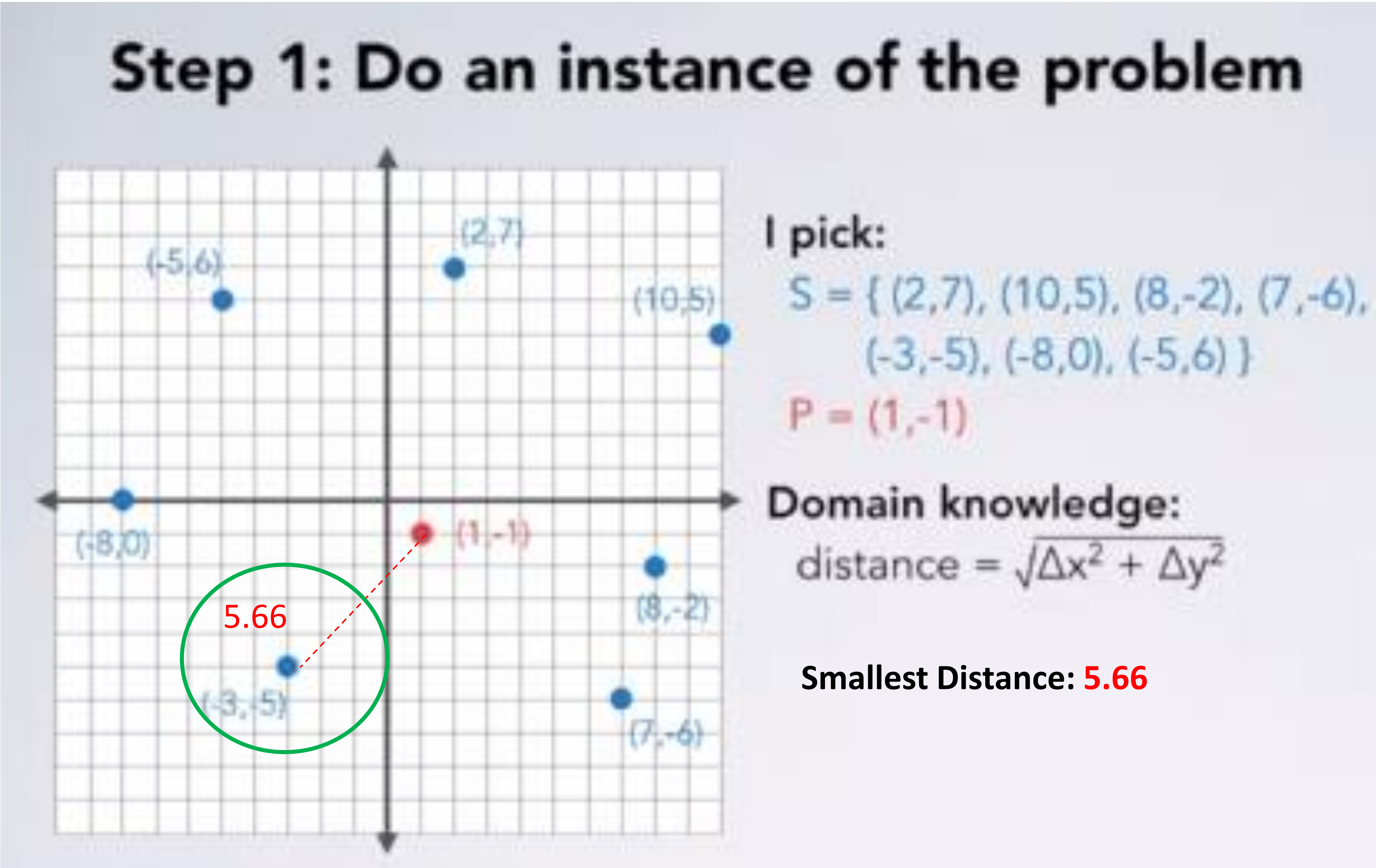
**Smallest Distance: 5.66**

# Example-2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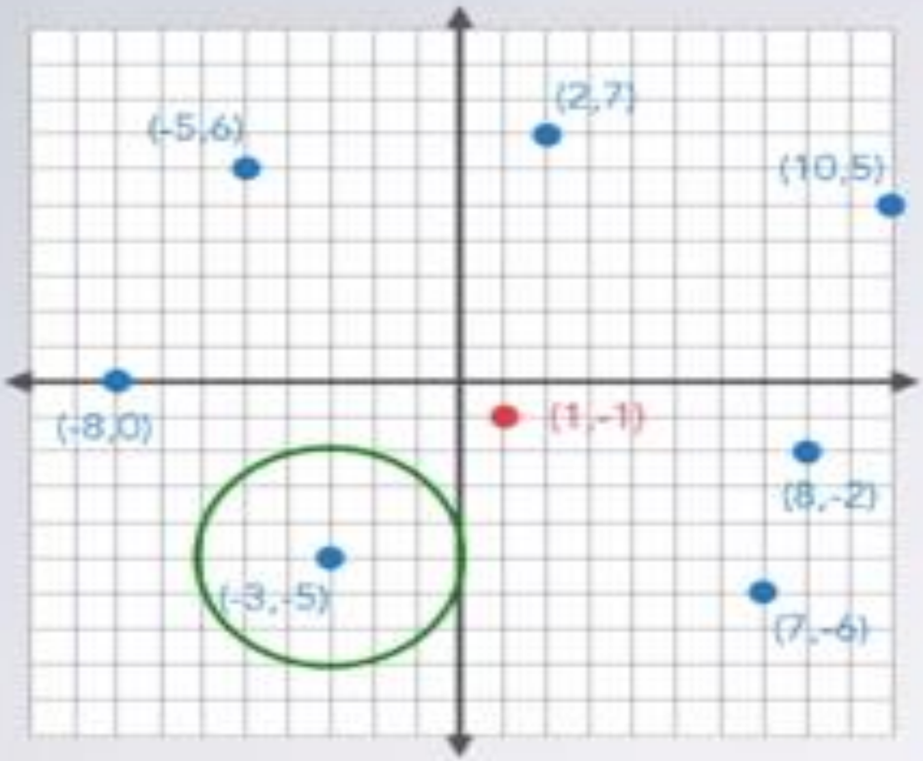
# Example-2

## Step 2: Write down what you just did




Computed $\sqrt{1^2 + 8^2} = 8.06$

Computed $\sqrt{9^2 + 6^2} = 10.82$

Compared 10.82 to 8.06—8.06 is smaller

Computed $\sqrt{7^2 + -1^2} = 7.07$

Compared 7.07 to 8.06—7.07 is smaller

Computed $\sqrt{6^2 + -5^2} = 7.81$

Compared 7.81 to 7.07—7.07 is smaller

Computed $\sqrt{-4^2 + -4^2} = 5.66$

Compared 5.66 to 7.07—5.66 is smaller

Computed $\sqrt{-9^2 + 1^2} = 9.06$

Compared 9.06 to 5.66—5.66 is smaller

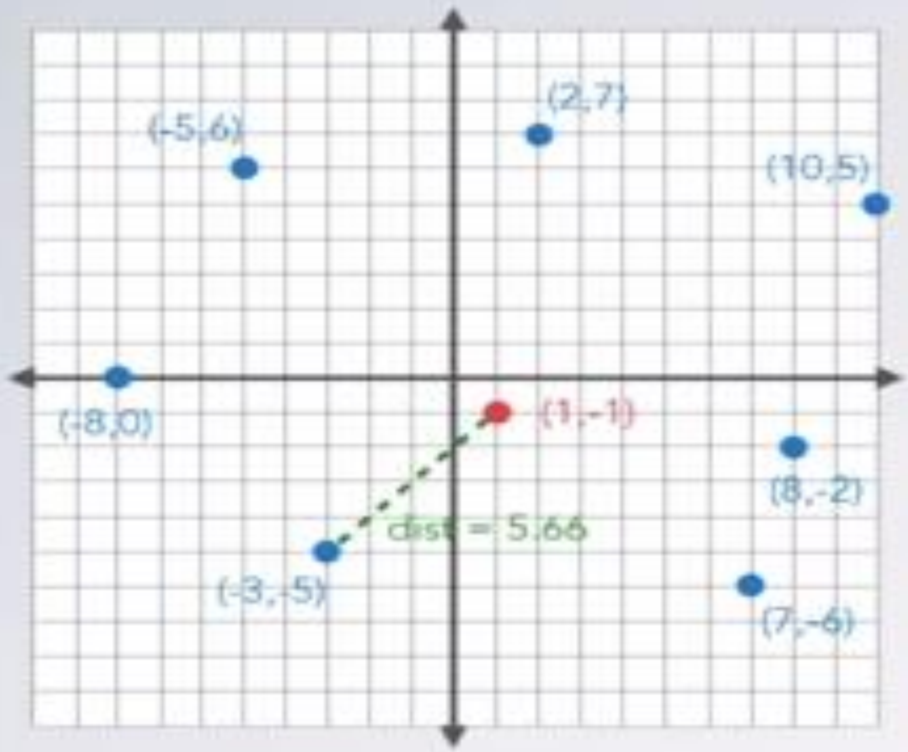
Computed $\sqrt{-6^2 + 7^2} = 9.22$

Compared 9.22 to 5.66—5.66 is smaller

I gave an answer of (-3,-5)

# Example-2

## Step 2: Write down what you just did




Computed $\sqrt{1^2 + 8^2} = 8.06$

**Started with best choice of (2,7)**

Computed $\sqrt{9^2 + 6^2} = 10.82$

Compared 10.82 to 8.06—8.06 is smaller

Computed $\sqrt{7^2 + -1^2} = 7.07$

Compared 7.07 to 8.06—7.07 is smaller

**Updated best choice to (8,-2)**

Computed $\sqrt{6^2 + -5^2} = 7.81$

Compared 7.81 to 7.07—7.07 is smaller

Computed $\sqrt{-4^2 + -4^2} = 5.66$

Compared 5.66 to 7.07—5.66 is smaller

**Updated best choice to (-3,-5)**

Computed $\sqrt{-9^2 + 1^2} = 9.06$

Compared 9.06 to 5.66—5.66 is smaller

Computed $\sqrt{-6^2 + 7^2} = 9.22$

Compared 9.22 to 5.66—5.66 is smaller

I gave an answer of (-3,-5)

# Example-2

## Step 3: Generalize these steps

Computed $\sqrt{1^2 + 8^2} = 8.06$
Started with best choice of (2,7)
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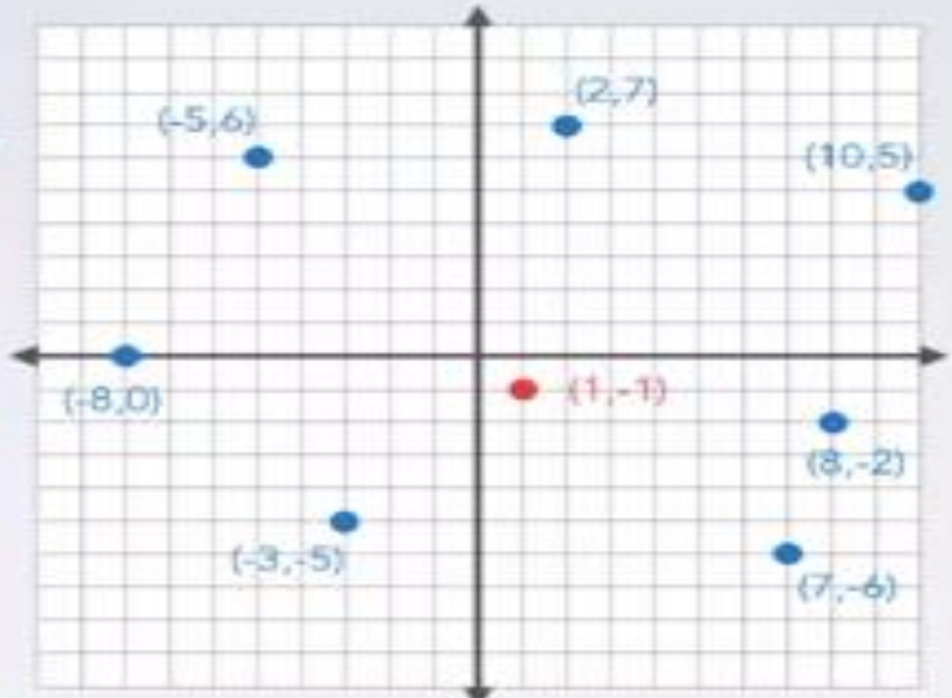
Compared 9.06 to 5.66—5.66 is smaller
Computed $\sqrt{-6^2 + 7^2} = 9.22$
Compared 9.22 to 5.66—5.66 is smaller
I gave an answer of (-3,-5)



# Example-2

## Step 3: Generalize these steps

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Compared 9.06 to 5.66—5.66 is smaller
Computed $\sqrt{-6^2 + 7^2} = 9.22$
Compared 9.22 to 5.66—5.66 is smaller
I gave an answer of (-3,-5)

We have some similar steps,
but we need to make them match up

# Example-2

## Step 3: Generalize these steps

Computed $\sqrt{1^2 + 8^2} = 8.06$
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**Updated best choice to (8,-2)**
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Compared 7.81 to 7.07—7.07 is smaller
Computed $\sqrt{-4^2 + -4^2} = 5.66$
Compared 5.66 to 7.07—5.66 is smaller
**Updated best choice to (-3,-5)**
Computed $\sqrt{-9^2 + 1^2} = 9.06$
Compared 9.06 to 5.66—5.66 is smaller
Computed $\sqrt{-6^2 + 7^2} = 9.22$
Compared 9.22 to 5.66—5.66 is smaller
I gave an answer of (-3,-5)

We have some similar steps,
but we need to make them match up

Some steps we do sometimes,
need to figure out condition

# Example-2

## Step 3: Generalize these steps

Computed $\sqrt{1^2 + 8^2} = 8.06$
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Compared 9.06 to 5.66—5.66 is smaller
Computed $\sqrt{-6^2 + 7^2} = 9.22$
Compared 9.22 to 5.66—5.66 is smaller
I gave an answer of (-3,-5)

We have some similar steps,
but we need to make them match up

Some steps we do sometimes,
need to figure out condition

Some steps we do at the start/end,
need to generalize specific numbers

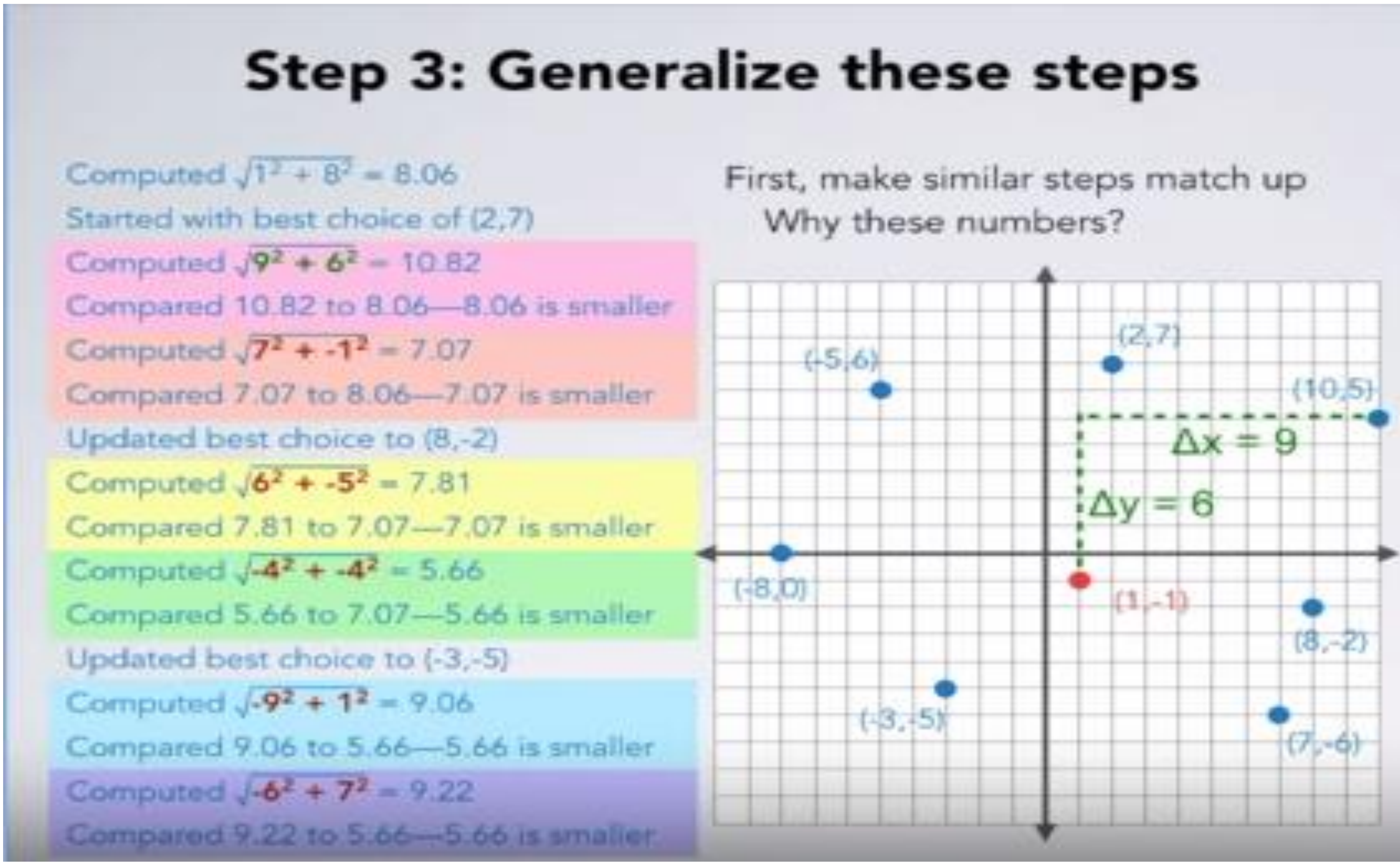
# Example-2

## Step 3: Generalize these steps

Computed $\sqrt{1^2 + 8^2} = 8.06$
Started with best choice of (2,7)
Computed $\sqrt{9^2 + 6^2} = 10.82$
Compared 10.82 to 8.06—8.06 is smaller
Computed $\sqrt{7^2 + -1^2} = 7.07$
Compared 7.07 to 8.06—7.07 is smaller
Updated best choice to (8,-2)
Computed $\sqrt{6^2 + -5^2} = 7.81$
Compared 7.81 to 7.07—7.07 is smaller
Computed $\sqrt{-4^2 + -4^2} = 5.66$
Compared 5.66 to 7.07—5.66 is smaller
Updated best choice to (-3,-5)
Computed $\sqrt{-9^2 + 1^2} = 9.06$
Compared 9.06 to 5.66—5.66 is smaller
Computed $\sqrt{-6^2 + 7^2} = 9.22$
Compared 9.22 to 5.66—5.66 is smaller
I gave an answer of (-3,-5)

First, make similar steps match up
Why these numbers?

# Example-2


Step 3: Generalize these steps
Computed $\sqrt{1^2 + 8^2} = 8.06$
Started with best choice of (2,7)
Computed $\sqrt{9^2 + 6^2} = 10.82$
Compared 10.82 to 8.06—8.06 is smaller
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Compared 5.66 to 7.07—5.66 is smaller
Updated best choice to (-3,-5)
Computed $\sqrt{-9^2 + 1^2} = 9.06$
Compared 9.06 to 5.66—5.66 is smaller
Computed $\sqrt{-6^2 + 7^2} = 9.22$
Compared 9.22 to 5.66—5.66 is smaller
First, make similar steps match up
Why these numbers?
(2,7)
(-5,6)
(10,5)
$\Delta x = 9$
$\Delta y = 6$
(-8,0)
(1,-1)
(8,-2)
(-3,-5)
(7,-6)

# Example-2

## Step 3: Generalize these steps

Computed $\sqrt{1^2 + 8^2} = 8.06$

Started with best choice of (2,7)

Computed $\sqrt{(S_1\text{'s } x - P\text{'s } x)^2 + (S_1\text{'s } y - P\text{'s } y)^2} = 10.82$

Compared 10.82 to 8.06—8.06 is smaller

Computed $\sqrt{(S_2\text{'s } x - P\text{'s } x)^2 + (S_2\text{'s } y - P\text{'s } y)^2} = 7.07$

Compared 7.07 to 8.06—7.07 is smaller

Updated best choice to (8,-2)

Computed $\sqrt{(S_3\text{'s } x - P\text{'s } x)^2 + (S_3\text{'s } y - P\text{'s } y)^2} = 7.81$

Compared 7.81 to 7.07—7.07 is smaller

Computed $\sqrt{(S_4\text{'s } x - P\text{'s } x)^2 + (S_4\text{'s } y - P\text{'s } y)^2} = 5.66$

Compared 5.66 to 7.07—5.66 is smaller

Updated best choice to (-3,-5)

Computed $\sqrt{(S_5\text{'s } x - P\text{'s } x)^2 + (S_5\text{'s } y - P\text{'s } y)^2} = 9.06$

Compared 9.06 to 5.66—5.66 is smaller

Computed $\sqrt{(S_6\text{'s } x - P\text{'s } x)^2 + (S_6\text{'s } y - P\text{'s } y)^2} = 9.22$

Compared 9.22 to 5.66—5.66 is smaller

I gave an answer of (-3,-5)

# Example-2

## Step 3: Generalize these steps

Computed $\sqrt{1^2 + 8^2} = 8.06$

Started with best choice of (2,7)

Computed $\sqrt{(S_1\text{'s } x - P\text{'s } x)^2 + (S_1\text{'s } y - P\text{'s } y)^2} = 10.82$

Compared 10.82 to 8.06—8.06 is smaller

Computed $\sqrt{(S_2\text{'s } x - P\text{'s } x)^2 + (S_2\text{'s } y - P\text{'s } y)^2} = 7.07$

Compared 7.07 to 8.06—7.07 is smaller

Updated best choice to (8,-2)

Computed $\sqrt{(S_3\text{'s } x - P\text{'s } x)^2 + (S_3\text{'s } y - P\text{'s } y)^2} = 7.81$

Compared 7.81 to 7.07—7.07 is smaller

Computed $\sqrt{(S_4\text{'s } x - P\text{'s } x)^2 + (S_4\text{'s } y - P\text{'s } y)^2} = 5.66$

Compared 5.66 to 7.07—5.66 is smaller

Updated best choice to (-3,-5)

Computed $\sqrt{(S_5\text{'s } x - P\text{'s } x)^2 + (S_5\text{'s } y - P\text{'s } y)^2} = 9.06$

Compared 9.06 to 5.66—5.66 is smaller

Computed $\sqrt{(S_6\text{'s } x - P\text{'s } x)^2 + (S_6\text{'s } y - P\text{'s } y)^2} = 9.22$

Compared 9.22 to 5.66—5.66 is smaller

I gave an answer of (-3,-5)

# Example-2

## Step 3: Generalize these steps

Computed $\sqrt{1^2 + 8^2} = 8.06$

Started with best choice of (2,7)

Computed $\sqrt{(S_1\text{'s } x - P\text{'s } x)^2 + (S_1\text{'s } y - P\text{'s } y)^2} =$ **10.82**

Compared 10.82 to 8.06—8.06 is smaller

Computed $\sqrt{(S_2\text{'s } x - P\text{'s } x)^2 + (S_2\text{'s } y - P\text{'s } y)^2} =$ **7.07**

Compared 7.07 to 8.06—7.07 is smaller

Updated best choice to (8,-2)

Computed $\sqrt{(S_3\text{'s } x - P\text{'s } x)^2 + (S_3\text{'s } y - P\text{'s } y)^2} =$ **7.81**

Compared 7.81 to 7.07—7.07 is smaller

Computed $\sqrt{(S_4\text{'s } x - P\text{'s } x)^2 + (S_4\text{'s } y - P\text{'s } y)^2} =$ **5.66**

Compared 5.66 to 7.07—5.66 is smaller

Updated best choice to (-3,-5)

Computed $\sqrt{(S_5\text{'s } x - P\text{'s } x)^2 + (S_5\text{'s } y - P\text{'s } y)^2} =$ **9.06**

Compared 9.06 to 5.66—5.66 is smaller

Computed $\sqrt{(S_6\text{'s } x - P\text{'s } x)^2 + (S_6\text{'s } y - P\text{'s } y)^2} =$ **9.22**

Compared 9.22 to 5.66—5.66 is smaller

I gave an answer of (-3,-5)

# Example-2

## Step 3: Generalize these steps

Computed $\sqrt{1^2 + 8^2} = 8.06$

Started with best choice of (2,7)

Computed $\sqrt{(S_1\text{'s } x - P\text{'s } x)^2 + (S_1\text{'s } y - P\text{'s } y)^2}$ call it **currentDistance**

Compared **currentDistance** to 8.06—8.06 is smaller

Computed $\sqrt{(S_2\text{'s } x - P\text{'s } x)^2 + (S_2\text{'s } y - P\text{'s } y)^2}$ call it **currentDistance**

Compared **currentDistance** to 8.06—**currentDistance** is smaller

Updated best choice to (8,-2)

Computed $\sqrt{(S_3\text{'s } x - P\text{'s } x)^2 + (S_3\text{'s } y - P\text{'s } y)^2}$ call it **currentDistance**

Compared **currentDistance** to 7.07—7.07 is smaller

Computed $\sqrt{(S_4\text{'s } x - P\text{'s } x)^2 + (S_4\text{'s } y - P\text{'s } y)^2}$ call it **currentDistance**

Compared **currentDistance** to 7.07—**currentDistance** is smaller

Updated best choice to (-3,-5)

Computed $\sqrt{(S_5\text{'s } x - P\text{'s } x)^2 + (S_5\text{'s } y - P\text{'s } y)^2}$ call it **currentDistance**

Compared **currentDistance** to 5.66—5.66 is smaller

Computed $\sqrt{(S_6\text{'s } x - P\text{'s } x)^2 + (S_6\text{'s } y - P\text{'s } y)^2}$ call it **currentDistance**

Compared **currentDistance** to 5.66—5.66 is smaller

I gave an answer of (-3,-5)

# Example-2

## Step 3: Generalize these steps

Call it bestDistance

Computed $\sqrt{1^2 + 8^2} = 8.06$
Started with best choice of (2,7)
Computed $\sqrt{(S_1\text{'s } x - P\text{'s } x)^2 + (S_1\text{'s } y - P\text{'s } y)^2}$ call it **currentDistance**
Compared **currentDistance** to 8.06—8.06 is smaller
Computed $\sqrt{(S_2\text{'s } x - P\text{'s } x)^2 + (S_2\text{'s } y - P\text{'s } y)^2}$ call it **currentDistance**
Compared **currentDistance** to 8.06—**currentDistance** is smaller
Updated best choice to (8,-2) and **bestDistance** to **currentDistance**
Computed $\sqrt{(S_3\text{'s } x - P\text{'s } x)^2 + (S_3\text{'s } y - P\text{'s } y)^2}$ call it **currentDistance**
Compared **currentDistance** to **bestDistance**—**bestDistance** is smaller
Computed $\sqrt{(S_4\text{'s } x - P\text{'s } x)^2 + (S_4\text{'s } y - P\text{'s } y)^2}$ call it **currentDistance**
Compared **currentDistance** to **bestDistance**—**currentDistance** is smaller
Updated best choice to (-3,-5) and **bestDistance** to **currentDistance**
Computed $\sqrt{(S_5\text{'s } x - P\text{'s } x)^2 + (S_5\text{'s } y - P\text{'s } y)^2}$ call it **currentDistance**
Compared **currentDistance** to **bestDistance**—**bestDistance** is smaller
Computed $\sqrt{(S_6\text{'s } x - P\text{'s } x)^2 + (S_6\text{'s } y - P\text{'s } y)^2}$ call it **currentDistance**
Compared **currentDistance** to **bestDistance**—**bestDistance** is smaller
I gave an answer of (-3,-5)

# Example-2

## Step 3: Generalize these steps

Computed $\sqrt{(S_0\text{'s } x - P\text{'s } x)^2 + (S_0\text{'s } y - P\text{'s } y)^2}$ call it **bestDistance**

Started with best choice of $\boxed{(2,7)}$ → Replace it with S0

Computed $\sqrt{(S_1\text{'s } x - P\text{'s } x)^2 + (S_1\text{'s } y - P\text{'s } y)^2}$ call it **currentDistance**

Compared **currentDistance** to **bestDistance**—**bestDistance** is smaller

Computed $\sqrt{(S_2\text{'s } x - P\text{'s } x)^2 + (S_2\text{'s } y - P\text{'s } y)^2}$ call it **currentDistance**

Compared **currentDistance** to **bestDistance**—**currentDistance** is smaller

Updated best choice to $\boxed{(8,-2)}$ and **bestDistance** to **currentDistance** → Replace it with S1

Computed $\sqrt{(S_3\text{'s } x - P\text{'s } x)^2 + (S_3\text{'s } y - P\text{'s } y)^2}$ call it **currentDistance**

Compared **currentDistance** to **bestDistance**—**bestDistance** is smaller

Computed $\sqrt{(S_4\text{'s } x - P\text{'s } x)^2 + (S_4\text{'s } y - P\text{'s } y)^2}$ call it **currentDistance**

Compared **currentDistance** to **bestDistance**—**currentDistance** is smaller

Updated best choice to $\boxed{(-3,-5)}$ and **bestDistance** to **currentDistance** → Replace it with S2

Computed $\sqrt{(S_5\text{'s } x - P\text{'s } x)^2 + (S_5\text{'s } y - P\text{'s } y)^2}$ call it **currentDistance**

Compared **currentDistance** to **bestDistance**—**bestDistance** is smaller

Computed $\sqrt{(S_6\text{'s } x - P\text{'s } x)^2 + (S_6\text{'s } y - P\text{'s } y)^2}$ call it **currentDistance**

Compared **currentDistance** to **bestDistance**—**bestDistance** is smaller

I gave an answer of (-3,-5)

# Example-2

Computed $\sqrt{(S_0\text{'s } x - P\text{'s } x)^2 + (S_0\text{'s } y - P\text{'s } y)^2}$ call it **bestDistance**

Started with best choice of $\mathbf{S_0}$

Computed $\sqrt{(S_1\text{'s } x - P\text{'s } x)^2 + (S_1\text{'s } y - P\text{'s } y)^2}$ call it **currentDistance**

If **currentDistance** is smaller than **bestDistance**

Then update **bestChoice** to $\mathbf{S_1}$ and **bestDistance** to **currentDistance**

Computed $\sqrt{(S_2\text{'s } x - P\text{'s } x)^2 + (S_2\text{'s } y - P\text{'s } y)^2}$ call it **currentDistance**

If **currentDistance** is smaller than **bestDistance**

Then update **bestChoice** to $\mathbf{S_2}$ and **bestDistance** to **currentDistance**

Computed $\sqrt{(S_3\text{'s } x - P\text{'s } x)^2 + (S_3\text{'s } y - P\text{'s } y)^2}$ call it **currentDistance**

If **currentDistance** is smaller than **bestDistance**

Then update **bestChoice** to $\mathbf{S_3}$ and **bestDistance** to **currentDistance**

Computed $\sqrt{(S_4\text{'s } x - P\text{'s } x)^2 + (S_4\text{'s } y - P\text{'s } y)^2}$ call it **currentDistance**

If **currentDistance** is smaller than **bestDistance**

Then update **bestChoice** to $\mathbf{S_4}$ and **bestDistance** to **currentDistance**

Computed $\sqrt{(S_5\text{'s } x - P\text{'s } x)^2 + (S_5\text{'s } y - P\text{'s } y)^2}$ call it **currentDistance**

If **currentDistance** is smaller than **bestDistance**

Then update **bestChoice** to $\mathbf{S_5}$ and **bestDistance** to **currentDistance**

Computed $\sqrt{(S_6\text{'s } x - P\text{'s } x)^2 + (S_6\text{'s } y - P\text{'s } y)^2}$ call it **currentDistance**

If **currentDistance** is smaller than **bestDistance**

Then update **bestChoice** to $\mathbf{S_6}$ and **bestDistance** to **currentDistance**

I gave an answer of (-3,-5)

# Example-2

## Step 3: Generalize these steps

Compute $\sqrt{(S_0\text{'s } x - P\text{'s } x)^2 + (S_0\text{'s } y - P\text{'s } y)^2}$ call it **bestDistance**

Start with best choice of $\mathbf{S_0}$

Count from 1 to the number of points in S exclusive, call each number **i**

- Compute $\sqrt{(S_i\text{'s } x - P\text{'s } x)^2 + (S_i\text{'s } y - P\text{'s } y)^2}$ call it **currentDistance**
- If **currentDistance** is smaller than **bestDistance**
  - Then update **bestChoice** to $\mathbf{S_i}$ and **bestDistance** to **currentDistance**

I gave an answer of (-3,-5)